

Notes: Womack (JF, 1996)

Do Brokerage Analysts' Recommendations Have Investment Value?

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	2
3.1	Recommendation-change groups	2
3.2	Return windows	3
3.3	Expected-return models	3
4	Models and empirical specifications	3
4.1	Market-model event returns	3
4.2	Size-adjusted and industry-adjusted returns	3
4.3	Fama-French abnormal returns	3
4.4	Approximate randomization	4
5	Identification and empirical design	4
6	Tables and figures	4
6.1	Table I: Description of Recommendation Change Samples; Table II: Financial Characteristics and Ratios for Recommendation Change Samples	4
6.2	Table III: Cumulative Returns Associated with Buy and Sell Recommendation Change Events; Figure 1: Short-run average excess returns and volume	5
6.3	Figure 2: Long-run average size-adjusted returns for buy and sell recommendation changes; Table IV: Consistency of the Event and Post-Event Returns	6
6.4	Figure 3: Long-run average size-adjusted returns for buy and sell recommendation changes stratified by size	7
6.5	Table V: Approximate Randomization Testing for Significance and Timing of Post-Event Returns	7
6.6	Table VI: Decomposition of Excess Returns into Stock Selection and Industry Selection Components	7
7	Short summary	9
8	Conclusion	10
8.1	Core conclusion	10
8.2	Incremental contribution revisited	10
8.3	Economic interpretation	10
8.4	Implications for market efficiency	10
8.5	Limitations	10
8.6	Takeaway	10

1 Big picture

- **Question.** Do brokerage analysts' recommendation changes contain investment value, and does the market fully incorporate this information immediately?
- **Setting.** The paper studies recommendation changes by fourteen major U.S. brokerage firms, focusing on added-to-buy, removed-from-buy, added-to-sell, and removed-from-sell revisions.
- **Main event result.** The three-day event reaction is large: added-to-buy recommendations earn positive abnormal returns and added-to-sell recommendations earn negative abnormal returns.
- **Main drift result.** The initial price response is incomplete. Buy recommendations have modest short-lived positive drift, while sell recommendations have larger negative drift lasting up to six months.
- **Asymmetry.** Sell recommendations are rarer and more informative than buy recommendations.
- **Skill result.** Randomization and decomposition tests suggest both stock-selection ability and market-timing ability.
- **Takeaway.** Brokerage recommendations have investment value, and prices underreact especially to negative analyst information.

2 Incremental contribution of the paper

- **Precise recommendation timing.** Earlier work often used published research dates that could lag the actual recommendation. Womack uses First Call recommendation dates that better capture when news reaches investors.
- **Complete recommendation-change taxonomy.** The paper analyzes not only new buy recommendations but also new sell recommendations and removals from buy and sell lists.
- **Sell-side evidence.** New sell recommendations were rarely studied before. The paper shows they produce large immediate declines and persistent negative drift.
- **Event plus drift.** The paper separates the immediate market reaction from post-event returns, showing both information content and incomplete adjustment.
- **Multiple benchmarks.** Size, industry, and Fama-French adjustments show the findings are not a simple expected-return artifact.
- **Stock selection versus timing.** Randomization tests and industry decompositions separate stock-picking information from market- or industry-timing effects.
- **Early analyst-value evidence.** The paper helped establish that analyst recommendations can be valuable even for large, heavily followed stocks.

3 Data and measurement

3.1 Recommendation-change groups

The sample contains 1,573 recommendation changes. The four groups are added-to-buy, removed-from-buy, added-to-sell, and removed-from-sell. The buy-related groups are much larger than the sell-related groups.

Interpretation: The small number of sell recommendations is meaningful. Analysts and brokerage firms issue sells less often, so an added-to-sell call may represent a more unusual and therefore more informative event.

3.2 Return windows

The paper measures returns before, during, and after the recommendation event:

- six-month and one-month pre-event returns;
- three-day recommendation event returns;
- one-, three-, and six-month post-event returns;
- total market impact, combining event returns and post-event drift.

Interpretation: The distinction between event-period returns and post-event returns is the core of the market-efficiency test. An efficient market may react at the event, but should not display systematic drift afterward.

3.3 Expected-return models

The paper uses several return benchmarks:

- unadjusted returns;
- CRSP size-decile adjusted returns;
- industry-adjusted returns based on four-digit SIC and size adjustment;
- Fama-French three-factor abnormal returns.

Interpretation: Robustness across benchmarks makes it harder to explain the return patterns as risk compensation or model misspecification.

4 Models and empirical specifications

4.1 Market-model event returns

For stock i on event day t :

$$ER_{i,t} = r_{i,t} - r_{m,t}, \quad ER_t = \frac{1}{n_t} \sum_{i=1}^{n_t} ER_{i,t}.$$

Interpretation: Event returns measure the immediate market response to the recommendation change.

4.2 Size-adjusted and industry-adjusted returns

$$ER_{i,t}^{size} = r_{i,t} - r_{size\ decile,t}.$$

Industry-adjusted returns subtract the average size-adjusted return of firms in the same four-digit SIC industry:

$$ER_{i,t}^{ind} = ER_{i,t}^{size} - \overline{ER}_{industry,t}^{size}.$$

Interpretation: These adjustments test whether abnormal performance is firm-specific rather than driven by firm size or industry selection.

4.3 Fama-French abnormal returns

$$ER_{i,t}^{FF} = r_{i,t} - r_{f,t} - \hat{\beta}_{1i}(r_{m,t} - r_{f,t}) - \hat{\beta}_{2i}SMB_t - \hat{\beta}_{3i}HML_t.$$

Interpretation: If abnormal returns survive this adjustment, they are unlikely to be explained by standard market, size, or value exposures.

4.4 Approximate randomization

The paper compares actual post-event returns with returns using the same firms but randomly shuffled event dates, random dates from the whole period, and market returns after actual recommendation dates.

Interpretation: This tests whether recommendation returns come from the analyst’s chosen stock, the analyst’s chosen date, or broad market conditions.

5 Identification and empirical design

- **Event-study design.** The key event is a recommendation change. The benchmark is expected stock performance absent the event.
- **Accurate dates.** First Call recommendation dates reduce the timing error that affected studies using later published research reports.
- **Underreaction test.** Post-event drift after a large event-window reaction indicates incomplete market adjustment.
- **Risk controls.** Size, industry, and Fama-French adjustments address different forms of expected-return misspecification.
- **Randomization validation.** Shuffled-date tests keep the same firms but break the link between analysts’ timing and returns.
- **Stock versus industry.** Industry-adjusted returns and Table VI decompose abnormal performance into stock-specific and industry components.
- **Main caveat.** The paper is an event-study, not a randomized experiment. It shows predictive investment value and market underreaction, not a guaranteed frictionless trading rule.

6 Tables and figures

6.1 Table I: Description of Recommendation Change Samples; Table II: Financial Characteristics and Ratios for Recommendation Change Samples

Read: Table I describes the four recommendation-change samples. Table II reports price, valuation, earnings forecast, prior return, book-to-price, and dividend-yield characteristics.

Economic message: The samples are mainly large-cap and well-followed firms. Finding post-event drift in these firms is especially important because information should be relatively abundant.

Interpretation: The tables also show the institutional asymmetry in analyst advice: sell-related recommendations are much rarer than buy-related recommendations.

Table I
Description of Recommendation Change Samples
Samples are obtained from searching the database, *First Call*, which is created and sold by First Call Corporation, Boston, Mass., to professor institutional investment managers. Observations are limited to those made by the top 14 brokerage research departments according to the 1989 *s* 1990 October issues of *Institutional Investor*.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Sample Category	Number of Obs. in Final Sample (on Separate Cos.)	Date Range of Sample	Number (%) in Sample in Market Cap. Deciles 9-10 (Large Cap.)	Number (%) in Sample in Market Cap. Deciles 6-8 (Medium Cap.)	Number (%) in Sample in Market Cap. Deciles 1-5 (Small Cap.)	Mean Number of Analysts With Earnings Per Share (EPS) Est. as Reported by I/B/E/S	Number (%) in Sam. w/ Concurr EPS Repo
Added-to-Buy- List changes	694 (528)	Jan. 1989- Jun. 1990	384 (55%)	221 (32%)	89 (13%)	17.9	64 (9%)
Removed-from- Buy-List changes	570 (420)	Jan. 1989- Jun. 1990	350 (61%)	182 (32%)	38 (7%)	19.2	84 (15%)
Added-to-Sell- List changes*	209* (185)	Jan. 1989- Dec. 1991	107 (51%)	73 (35%)	29 (14%)	17.8	30 (15%)
Removed-from- Sell-List changes	100 (95)	Jan. 1989- Jun. 1990	54 (54%)	38 (38%)	8 (8%)	18.9	13 (13%)
Total Observations	1573		895 (57%)	514 (33%)	164 (10%)	18.4	191 (12%)

* After preliminary analysis, additional observations were collected for the added-to-sell category. Thus, the annualized ratio of buy-to- recommendations is closer to 7 to 1 rather than 7 to 2 as shown here.

Table II
Financial Characteristics and Ratios for Recommendation Change Samples

This table lists financial characteristics and ratios of the samples described in Table I. Median values of the financial measures are presented in *n* cases. Means are similar but skewed right, especially for PE ratios. The 10%-90% range is given in some cases to describe the location of the cen 80 percent of the values for that characteristic.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Sample Category	Median Stock Price Per Share (Range)	Median PE Ratio at Event Time Using I/B/E/S Est. for Most Current Year	Median PE Ratio at Event Time Using Trailing 12 Mos. (Realized) EPS	Percent Change Current Yr. Mean I/B/E/S EPS Est. Mo. Before to Mo. After Event	Size-Adj. (Excess) Return for 6-Mo. Period Prior to Event (<i>t</i> -statistic)	Median (Mean) Price-to-Book Value at First Prior EPS Date	Media Divisor Yield at 1 of Ever Date Mo
Added-to-Buy- List changes	\$29.40 (10.5-63.4)	12.5x	13.6x	+2.7%	+1.2% (1.38)	1.8x (2.4x)	1.65% (0.0-4.6)
Removed-from-Buy- List changes	\$34.20 (13.5-67.0)	13.0x	13.5x	-2.9%	+5.0% (4.64)	1.9x (2.6x)	1.89% (0.0-4.7)
Added-to-Sell- List changes	\$25.00 (6.8-61.6)	12.9x	13.6x	-5.8%	-2.1% (-1.13)	1.6x (2.1x)	2.11% (0.0-6.6)
Removed-from-Sell- List changes	\$30.20 (12.0-63.8)	12.0x	13.4x	+3.1%	-5.7% (-2.73)	1.5x (2.1x)	2.91% (0.0-7.0)
Averages, All Observations	\$31.25	12.7x	13.5x	-0.3%	+0.3%	1.7x (2.4x)	.50%

(a) Table I: sample composition and analyst coverage.

(b) Table II: financial characteristics by recommendation group.

Descriptive tables are paired to compare sample construction with firm characteristics.

6.2 Table III: Cumulative Returns Associated with Buy and Sell Recommendation Change Events; Figure 1: Short-run average excess returns and volume

Read: Table III reports immediate and longer-window returns under several return adjustments. Figure 1 plots event-time price relatives and abnormal volume for added-to-buy and added-to-sell calls.

Economic message: The event-window reaction is large: positive for added-to-buy and negative for added-to-sell. Trading volume spikes at the event, showing that investors pay attention to analyst recommendation changes.

Interpretation: The sell recommendation effect is especially revealing because sells are rare. The market treats new sells as unusually negative signals.

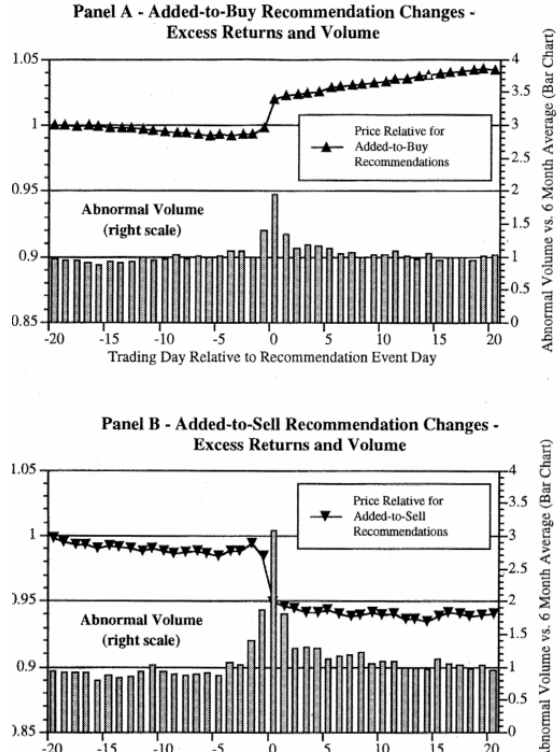
Table III
Cumulative Returns Associated with Buy and Sell Recommendation Change Events

Actual, size-adjusted (using Center for Research in Security Prices (CRSP) value-weighted index deciles), industry-adjusted, and Fama/French model returns for various periods surrounding buy and sell recommendation changes. For the first three column sets, the event is the three-day window centered on the First Call database event date. For the Fama and French (1993) three-factor model, the event is the calendar month in which the recommendation takes place. Standard deviations (SDs) and t-statistics are cross-sectional, calculated from all companies with returns in that period.

	Actual Returns (unadjusted)		Size-Adjusted Returns (CRSP VW Index Deciles)			Industry-Adjusted Returns (4-Digit SIC & CRSP VW Deciles)			Excess Returns (Fama-French 3-Factor Model)		
	Mean	S.D.	Mean	S.D.	t-stat.	Mean	S.D.	t-stat.	Mean	S.D.	t-stat.
Measurement Periods Relative to 3-Day Event Window											
Panel A: Added-to-Buy-List Recommendation Changes											
	(N=694)					(N=464)					
6-Month Period Before Event	6.24	23.75	1.21	22.00	1.38	2.84	24.30	2.47	-Not Available-		
1-Month Period Before Event	0.65	10.64	0.04	9.46	0.12	0.29	9.61	0.66	-Not Available-		
Recommendation 3-Day Event	3.27	5.64	2.96	5.47	14.06	2.84	5.41	11.31	4.00*	9.94	9.77
1st Month After Event	4.57	8.40	2.38	7.91	7.75	2.33	9.45	5.32	1.87*	8.23	5.52
3-Month Period After Event	4.38	16.93	1.75	15.41	2.92	2.44	16.22	3.24	2.12*	15.73	3.27
6-Month Period After Event	-0.86	26.35	0.09	24.80	0.09	3.73	26.16	3.07	2.42*	25.62	2.38
Total Mkt. Impact/Event+1 Mo.	7.97	10.22	5.42	9.68	14.43	5.23	11.01	10.23	5.93*	13.11	10.96
Panel B: Removed-from-Buy-List Recommendation Changes											
	(N=570)					(N=383)					
6-Month Period Before Event	12.30	27.21	5.02	25.31	4.64	4.58	24.72	3.62	-Not Available-		
1-Month Period Before Event	3.65	10.32	1.54	9.97	3.64	1.25	11.28	2.18	-Not Available-		
Recommendation 3-Day Event	-1.62	6.00	-1.84	5.58	-8.17	-2.08	5.97	-6.82	-1.61*	10.33	-3.52
1st Month After Event	0.82	8.92	-1.02	8.20	-2.93	-0.93	9.03	-2.02	-0.01*	9.18	-0.01
3-Month Period After Event	-0.68	16.75	-2.58	14.51	-4.17	-1.31	15.35	-1.67	-0.64*	15.00	-0.96
6-Month Period After Event	-4.36	25.01	-5.50	21.78	-5.93	-2.19	21.96	-1.95	-3.28*	22.89	-2.34
Total Mkt. Impact/Event+6 Mo.	-5.70	25.45	-7.15	22.24	-7.52	-4.03	22.60	-3.48	-3.37*	26.29	-2.88

(a) Table III, part 1: added-to-buy and removed-from-buy returns.

Immediate price and volume reactions. The table provides estimates, while the figure shows the event-time pattern.



(b) Figure 1: short-run returns and volume.

Panel C: Added-to-Sell-List Recommendation Changes											
			(N=209)			(N=139)					
6-Month Period Before Event	4.38	31.85	-2.13	26.91	-1.13	-4.85	24.42	-2.31	-Not Available-		
1-Month Period Before Event	0.76	14.53	-0.32	13.87	-0.33	-0.23	14.71	-0.18	-Not Available-		
Recommendation 3-Day Event	-4.32	10.01	-4.69	9.96	-6.78	-4.99	10.63	-5.53	-3.87*	15.03	-
1st Month After Event	0.45	13.12	-0.73	12.17	-0.87	0.38	11.46	0.39	-1.51*	12.99	-
3-Month Period After Event	-2.32	21.42	-5.41	19.46	-4.00	-4.15	17.34	-2.81	-5.91*	19.93	-
6-Month Period After Event	-4.43	30.13	-9.15	25.24	-5.14	-5.65	24.79	-2.65	-6.42*	28.02	-
Total Mkt. Impact/Event+6 Mo.	-8.39	30.69	-13.71	26.21	-7.41	-10.38	27.29	-4.41	-11.14*	28.43	-

Panel D: Removed-from-Sell-List Recommendation Changes											
			(N=100)			(N=62)					
6-Month Period Before Event	-0.02	22.39	-5.69	20.29	-2.73	-9.69	20.16	-3.75	-Not Available-		
1-Month Period Before Event	0.28	11.99	-1.24	10.76	-1.13	-0.82	12.58	-0.51	-Not Available-		
Recommendation 3-Day Event	0.71	6.45	0.32	6.43	0.50	-0.37	7.48	-0.39	1.42*	11.15	-
1st Month After Event	2.82	10.36	1.02	9.99	1.00	-0.40	9.68	-0.32	0.93*	8.00	-
3-Month Period After Event	-0.56	16.21	-1.50	15.13	-0.97	-4.04	26.11	-1.21	-1.82*	14.27	-
6-Month Period After Event	-8.15	23.49	-5.57	21.39	-2.54	-3.27	30.48	-0.84	-4.78*	19.98	-
Total Mkt. Impact/Event+6 Mo.	-7.66	23.89	-5.35	21.79	-2.39	-3.68	30.58	-0.94	-2.94*	22.73	-

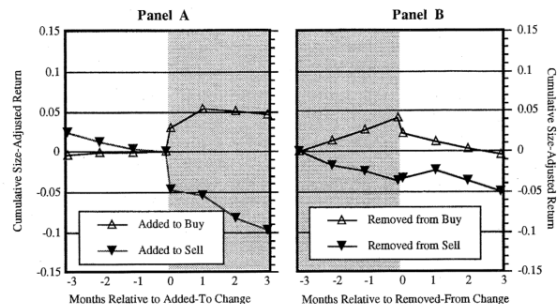
Table III, part 2: added-to-sell and removed-from-sell recommendation changes.

6.3 Figure 2: Long-run average size-adjusted returns for buy and sell recommendation changes; Table IV: Consistency of the Event and Post-Event Returns

Read: Figure 2 shows long-run cumulative size-adjusted returns. Table IV reports mean, median, and subperiod event and post-event returns.

Economic message: The post-event drift is strongest and most persistent for added-to-sell recommendations. Buy recommendations drift modestly upward for about one month.

Interpretation: This pair shows the central inefficiency result: the market reacts immediately but not completely.



Long-run average size-adjusted returns for buy and sell recommendation changes
average cumulative size-adjusted returns for the four types of recommendation changes stratified by size

(a) Figure 2: long-run size-adjusted returns.

Long-run drift and subperiod consistency. The figure gives the return path; the table verifies its magnitude and persistence.

Table IV
Consistency of the Event and Post-Event Returns: Means, Medians, and Subperiod Returns

Mean size-adjusted returns are presented for overall samples and for the three subperiods of the sample time frame. Overall median returns are also shown. Asterisks denote significance at $\alpha = 0.05$.

	Three-Day Event Return	One Month Postevent Return	Six-Month Postevent Return
Panel A: Added-to-Buy-List Recommendation Changes			
Mean, all observations	3.0*	2.4*	1.8
Jan/89–Jun/89	2.7*	1.6*	4.3
Jul/89–Dec/89	3.3*	2.1*	3.3
Jan/90–Jun/90	2.7*	2.8*	–3.9
Median, all observations	2.0	1.8	0.2
Panel B: Added-to-Sell-List Recommendation Changes			
Mean, all observations	–4.7*	–0.7	–9.1*
Jan/89–Dec/89	–2.7*	–0.3	–11.4*
Jan/90–Dec/90	–5.6*	–1.7	–11.6*
Jan/91–Dec/91	–5.3*	0.1	–4.2 (t = –1.8)
Median, all observations	–2.8	–0.6	–7.3

(b) Table IV: consistency across subperiods.

6.4 Figure 3: Long-run average size-adjusted returns for buy and sell recommendation changes stratified by size

Read: Figure 3 splits added-to-buy and added-to-sell recommendations by market capitalization decile group: small, medium, and large.

Economic message: Drift is strongest for small firms. This is consistent with slower information diffusion where information processing costs, attention constraints, or trading frictions are larger.

Interpretation: The size-stratified result strengthens the underreaction interpretation. Small-firm sell calls show especially large downward drift.

6.5 Table V: Approximate Randomization Testing for Significance and Timing of Post-Event Returns

Read: Table V compares actual post-event returns with returns from random and shuffled dates.

Economic message: Actual analyst-selected dates matter. The positive buy drift and negative sell drift are not replicated by randomly chosen dates for the same firms.

Interpretation: The table decomposes analysts' ability into stock picking and market timing. Analysts seem to identify both stocks and moments when returns are likely to move in the recommendation direction.

6.6 Table VI: Decomposition of Excess Returns into Stock Selection and Industry Selection Components

Read: Table VI decomposes size-adjusted excess returns into stock-selection and industry-selection components.

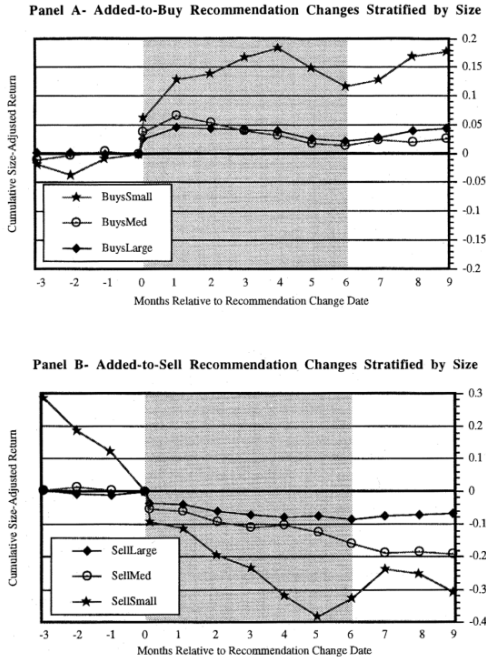


Figure 3: long-run average size-adjusted returns stratified by market capitalization size.

Table V
Approximate Randomization Testing for Significance and Timing of Post-Event Returns

Description of the Testing Procedure: First, the post-event returns (the "test statistic") for the actual added-to-buy list ("buy") and added-to-sell list ("sell") samples are calculated. For "buy" recommendations, it is the one-month (21 trading days) unadjusted buy-and-hold return, commencing two days after the recommendation. For "sell" recommendations, it is the six-month unadjusted return, also beginning two days after the recommendation. These actual post-event returns are shown in Panels A and B, Line 1. Second, using the same "buy" and "sell" sample firms, the event dates are randomly shuffled and the mean postevent (pseudo) return is compared to the actual post-event return. Column 3 of Line 2 in each panel represents the mean of 300 random sample mean returns generated in this way. Column 4 of Line 2 in each panel shows the number of random trials that exceeds the test statistic (NGE) shown in Line 1 in each panel. NGE means the Number Greater or Equal to the test statistic (the mean return in Line 1). For example, in Panel A, zero of 300 randomly generated (shuffled or randomly picked) mean returns exceeded the actual postevent return of +4.57 percent. However, in Panel B, 2 of 300 randomly shuffled sample means were less than the actual post-event "sell" return of -4.78 percent. Randomly generated dates (Lines 3 and 4) from the entire sample period are used to compare an average one-month or six-month return in the entire period sampled for the buy and sell samples and the average value-weighted market return over the same periods. Line 5 in each panel shows the mean of value-weighted market returns beginning two days after the actual event dates.

Calculation of Significance Levels: The ratio of the NGE to the number of simulations (300) shows the frequency with which the randomly generated portfolio mean return is greater (less) than the original buy (sell) test statistic. The significance level of the test, however, is the ratio $(NGE + 1)/(300 + 1)$. This minor adjustment insures that the test is valid (see Noreen, 1989). In the table headings, α is the rejection level of the test, and β is the confidence level of the test that is $\beta = \text{prob}(\phi < \alpha | \text{NGE, NS} | \text{uniform prior})$ where ϕ is the limit of $(NGE + 1)/(NS + 1)$ as NS approaches infinity. Panels A and B show that the null hypothesis that the postevent returns for recommendations are independent of the dates of the recommendations is rejected at $\alpha = .01$ for "buys" and $\alpha = .05$ for "sells."

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Sample Tested	Evaluated From	Mean Port. Return	NGE/Repl. cations	Significance Ratio	β Confidence Level @ $\alpha = 0.01$	$\alpha = 0.05$
Panel A: One-Month Post-Event Mean Return for Added-to-Buy-List Changes ($N = 694$)						
1. "Buy" changes	"Buy" Dates	+4.57%	NM			
2. "Buy" changes	"Buy" Dates Shuffled	+3.27%	0 of 300	0.0033	0.951	1.000
3. "Buy" changes	Random Dates in Period	+1.41%	0 of 300	0.0033	0.951	1.000
4. Market (Value-Weighted CRSP Index)	Random Dates in Period	+1.46%	0 of 300	0.0033	0.951	1.000
5. Market (VW average)	"Buy" Dates	+2.60%	NM			
Panel B: Six-Month Post-Event Mean Return for Added-to-Sell-List Changes ($N = 201$)						
1. "Sell" changes	"Sell" Dates	-4.78%	NM			
2. "Sell" changes	"Sell" Dates Shuffled	-0.48%	2 of 300	0.0069	0.578	1.000
3. "Sell" changes	Random Dates in Period	+3.05%	0 of 300	0.0033	0.951	1.000
4. Market (Value-Weighted CRSP Index)	Random Dates in Period	+6.99%	0 of 300	0.0033	0.951	1.000
5. Market (VW average)	"Sell" Dates	+4.32%	NM			

Table V: approximate randomization tests for post-event returns and timing.

Economic message: Analysts are not only selecting industries. A meaningful part of the abnormal performance comes from firm-specific stock selection.

Interpretation: This supports the interpretation that brokerage research produces firm-level information or judgment, not only broad sector timing.

Table VI
Decomposition of Excess Returns into Stock Selection and Industry Selection Components

Industry-adjusted, size-adjusted, and the difference between size- and industry-adjusted returns for all recommendation change observations with available industry-adjusted returns. Industry-adjusted returns (Column 1) are available for approximately 66 percent of sample observations (see Section II.A). Therefore, size-adjusted returns (Column 3) are similar but not identical to mean returns in Table III due to different sample sizes. Cross-sectional standard deviations and t -statistics are shown in Table III.

	Column 1 Industry-Adj. Rets. (4-Digit SIC & CRSP Deciles) "Stock Selection" Mean	Column 2 The Difference + "Industry Selection" Mean	Column 3 = Size-Adjusted Rets. (CRSP Size Adj. Returns) = Total Excess Return Mean
Panel A: Added-to-Buy-List Changes ($N = 464$)			
Recommendation 3-Day Event	2.84*	0.07	2.91*
1st Month After Event	2.33*	-0.29	2.04*
6-Month Period After Event**	3.73*	-4.26*-Inaccurate	-0.62
Panel B: Removed-from-Buy-List Changes ($N = 383$)			
Recommendation 3-Day Event	-2.08*	0.11	-1.96*
1st Month After Event	-0.93	-0.64	-1.58*
6-Month Period After Event	-2.19*	-4.22*-Accurate	-6.20*
Panel C: Added-to-Sell-List Changes ($N = 139$)			
Recommendation 3-Day Event	-5.00*	-0.25	-5.25*
1st Month After Event	0.38	-0.39	-0.01
6-Month Period After Event	-5.65*	-4.83*-Accurate	-10.23*
Panel D: Removed-from-Sell-List Changes ($N = 62$)			
Recommendation 3-Day Event	-0.37	0.17	-0.19
1st Month After Event	-0.79	-0.15	-0.55
6-Month Period After Event	-3.27	-2.22-Inaccurate,NS	-5.46

* Asterisk denotes significance (adjusted return $\neq 0$) at $\alpha = 0.01$.

** The 6-month period after event includes the first month after event.

"Accurate" denotes an "industry selection" component in the direction predicted by the analyst's recommendation.

NS = Not significant at $\alpha = 0.01$.

Table VI: decomposition into stock-selection and industry-selection components.

7 Short summary

- Womack studies recommendation changes by major U.S. brokerage firms using relatively precise First Call dates.
- Buy and sell recommendation changes produce large immediate price reactions.
- The market response is incomplete, especially for sell recommendations.
- Sell recommendations are rarer but more powerful and are followed by large negative drift.
- Results are robust to size, industry, and Fama-French adjustments.
- Randomization tests indicate both stock-selection and market-timing ability.
- Small-firm drift suggests information diffuses more slowly where attention and trading frictions are larger.

8 Conclusion

8.1 Core conclusion

Womack shows that brokerage analysts' recommendation changes have investment value. Prices react sharply on the event date but continue to drift afterward, especially after new sell recommendations.

8.2 Incremental contribution revisited

The paper's key contribution is to use precise recommendation-change dates, include sell recommendations and removals from lists, and separate announcement effects from post-event drift. This permits a much clearer measurement of analyst information and market underreaction than earlier studies.

8.3 Economic interpretation

Analysts may possess superior information, better interpretation of public signals, or credibility that focuses investor attention. New sell recommendations are especially informative because sell calls are rare and potentially costly for analysts to issue.

8.4 Implications for market efficiency

The results reject immediate full adjustment to recommendation changes. The market reacts, but not enough. The strongest underreaction appears for bad news, consistent with limits to attention, slower diffusion, or short-sale frictions.

8.5 Limitations

The sample covers major brokerage firms in the late 1980s and early 1990s. Trading costs, access to recommendations, changes in recommendation status, and short-sale constraints affect implementability. The results document investment value and underreaction, not a costless trading strategy for every investor.

8.6 Takeaway

Brokerage recommendations are economically meaningful signals. They move prices immediately and predict post-event returns, with the strongest evidence after sell recommendations.